3 Experiments

3.1 Dataset and Task

We use PRAGMEGA (Floyd, 2022; Hu et al., 2023) as a pragmatic reasoning dataset and task. This dataset covers seven broad pragmatic phenomena observed in English conversations. Each instance in the dataset requires correctly answering questions designed primarily to test understanding an utterance's implied meanings, considering the utterance itself and its conversational context.

Each problem in the dataset is classified into one of the seven pragmatic phenomena the dataset addresses. These seven pragmatic phenomena are *Deceits*, *Indirect speech*, *Irony*, *Maxims*, *Metaphor*, *Humor*, and *Coherence*. Since the present study specifically focuses on the task of correctly interpreting non-literal meanings in utterances, we conducted experiments targeting five of these pragmatic phenomena, excluding *Humor* and *Coherence*, which do not directly address such meanings. The following descriptions are provided in (Hu et al., 2023) for each phenomenon[1]:

- ***Deceits***: *Polite deceits* used for social or personal relationships. In the questions, respondents must employ *Theory of Mind* and other reasoning skills to determine why the speaker used a particular utterance correctly.
- ***Indirect Speech***: Utterances with performative meanings, such as prompting others to take action. The questions test whether respondents understand what the speaker tries to convey in stories involving indirect requests.
- ***Irony***: Utterances that communicate the opposite of their literal meaning. The questions require interpreting what the characters intend to convey through the presented irony.
- ***Maxims***: Utterances that violate one of Grice's four maxims. Respondents determine why the characters made such utterances.
- ***Metaphor***: Utterances that depict comparisons between entities in a non-literal manner. The questions present metaphors within a story and ask respondents to interpret what the speaker is trying to convey.

The total number of problems is 520, comprising 100 instances each for *Deceits*, *Indirect Speech*, and *Metaphor*, 125 instances for *Irony*, and 95 instances for *Maxims*. Our experiments do not explicitly indicate the phenomenon to which each problem belongs within the prompt. Instead, a common prompt format is used across all phenomena to present the problem and response structure.

[1]Examples of problems corresponding to each phenomenon are shown in Table 2 in the appendix.

3.2 Methods

We conducted comparative experiments on the following prompting methods (two baselines and two proposed methods). Baseline-1/2 denote baseline methods, and Proposed-1/2 denote proposed methods. The baseline and proposed methods are instance-agnostic, meaning they do not provide models with instance-dependent information or top-down hints. Within these instance-agnostic methods, we hypothesize that our proposed methods, including summaries of pragmatic theory in prompts, would achieve higher performance on pragmatic reasoning tasks than the baseline methods. This is because, by indicating these theories, we expect the model to know which area of knowledge or reasoning contained in the model should be called. Note that all experiments adopt a zero-shot learning setup, meaning no task-solving examples are provided in the prompts.

Baseline-1: Simple A setting in which the prompts explicitly instruct the models to output **only** the final selected answer.
Baseline-2: Chain-of-Thought (`cot`) A setting in which the model is prompted with "`Firstly, think step-by-step and write down your process of thinking,`" explicitly instructing it to output its reasoning process leading to the final answer, followed by the selected answer. This method is often referred to as *zero-shot Chain-of-Thought* (Kojima et al., 2022)[a].
Proposed-1: Gricean Prompting (`grice`) A setting in which the prompt provides the models with a brief overview of Gricean theory (Grice, 1989), explicitly instructing them to output a reasoning process aligned with this overview, followed by the final selected answer.
Proposed-2: Relevance Theory Prompting (`relevance`) A setting in which the prompt provides the models with a brief overview of Relevance Theory (Sperbel and Wilson, 1995; Carston, 2002), explicitly

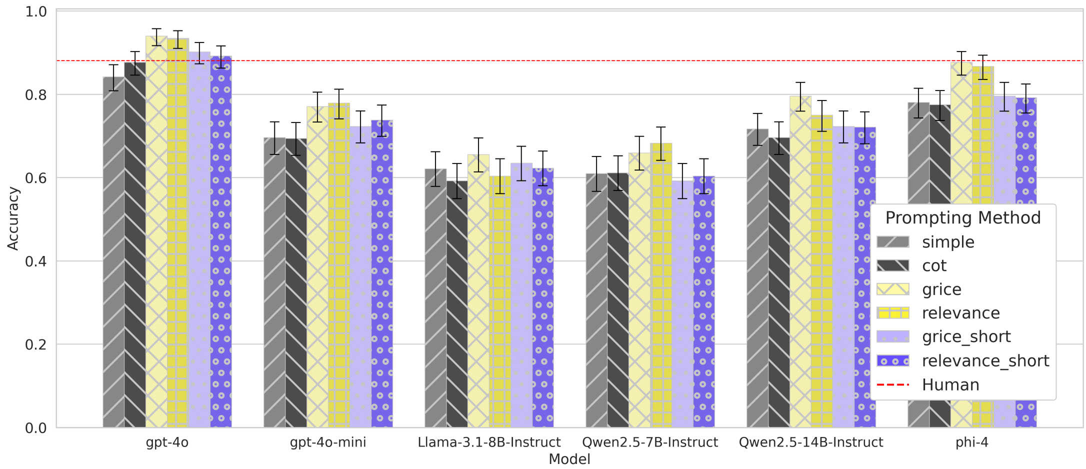

Figure 1: Accuracies on pragmatic inference task of PRAGMEGA. In most models, the proposed methods outperformed the baseline methods. The human scores indicate scores presented in the original paper by (Hu et al., 2023). Error bars represent 95% confidence intervals calculated using Wilson's method (Wilson, 1927). Even with a short prompt for the pragmatic theory, larger models showed improvements from the proposed methods; however, the extent of improvement was smaller compared to when the theory was explained in detail.

instructing them to describe a reasoning process aligned with this overview before outputting the final selected answer.

[a] Note that this method differs from the "Chain-of-Thought" approach presented in (Yerukola et al., 2024) and (Kim et al., 2023), as it does not provide instance-specific hints for correct interpretation.

3.3 Models

We conducted experiments using various LLMs implementing decoder-based Transformer architecture (Vaswani et al., 2017; Liu et al., 2018). We experimented with publicly available models (open models), whose source code and pre-trained parameters are released, and proprietary models (closed models), whose parameters are not publicly available. As Open models, we selected the LLaMa3 series (Grattafiori et al., 2024) and the Qwen2.5 series (Yang et al., 2025). As Closed models, we selected GPT-4o and GPT-4o mini (OpenAI, 2025). Considering the length of our prompts, we chose models with sufficiently large context lengths (16k tokens or more) for the experiments. Appendix §E shows the hyperparameters used in the experiments. About computation details, see §F in the Appendix.

4 Results

4.1 Experimental Results

The experimental results shown in Figure 1 confirm that the proposed methods perform better than the baseline methods on the pragmatic reasoning task[2]. Notably, with GPT-4o, the proposed methods enabled the model to achieve performance surpassing that of humans. Even in the case of phi-4, where there was a performance gap between the baseline methods and human scores, applying our methods allowed the model to reach human-level accuracy.

Using **Gricean prompting** (`grice`) resulted in higher scores than both baseline methods across all models tested. This method had the most significant effect on phi-4, where accuracy improved by 0.096 compared to the higher-scoring baseline method (`simple`). Even for Llama-3.1-8B-Instruct, where the most minor improvement was observed, accuracy increased by 0.032.

For **Relevance prompting** (`relevance`), no performance improvement was observed for Llama-3.1-8B-Instruct compared to the baseline, but all other models showed performance gains. When comparing `grice` and `relevance`, `grice` achieved higher scores in most models. While some models recorded higher scores with `relevance`, the difference from `grice` in those cases was minimal.

For **Short prompting** (`grice short`, `relevance short`), performance improvements over the baseline were observed in all models except Qwen2.5-7B-Instruct. Even in models where scores improved over the baseline, the magnitude of improvement was generally smaller than that achieved by the proposed methods.

[2] The exact scores are presented in §G, I in the Appendix.

4.2 Analysis

4.2.1 General Overview

When comparing `grice` and `relevance`, `grice` achieved higher scores in most models. While some models recorded higher scores with `relevance`, the difference from `grice` in those cases was minimal. However, this difference does not necessarily indicate that Gricean Theory is more valid than Relevance Theory. Instead, we speculate that the difference is due to Gricean Theory appearing more frequently in the training corpus of the models.

There was no clear trend indicating that longer input or output lengths consistently led to higher accuracy. An analysis of the relationships between input/output length and accuracy revealed little correlation between these factors. The Pearson correlation coefficients between accuracy and input length and accuracy and output length were 0.181 and 0.211, respectively. Additionally, the coefficient of determination (R^2) from simple regression analysis was 0.032 and 0.044, respectively (see Appendix §H for detailed analysis results).

4.2.2 Analyses by Pragmatic Phenomena

A summary of the analysis results for each of the five pragmatic phenomena included in PRAGMEGA, comparing different prompting methods using GPT-4o and Qwen2.5-7B-Instruct, is shown in Figure 2. [3] The results showed the most notable score improvement from the proposed methods in *Irony*. In the case of GPT-4o, baseline methods resulted in performance lower than that of humans; however, by applying the proposed methods, the model achieved human-level accuracy. For this phenomenon, regardless of whether the prompts provide the models with an overview of the theory, using Gricean Theory generally resulted in higher scores than Relevance Theory. In *Indirect Speech*, although the margin was smaller than in *Irony*, consistent improvements over the baseline methods were also observed. However, we found no clear superiority between Gricean Theory and Relevance Theory in this phenomenon. For *Maxims*, using Gricean Prompting, which includes an overview of Grice's maxims, did not always lead to improved scores over the baseline. However, for models with 14B or more parameters, the proposed methods generally resulted in higher accuracy. In most cases, models performed better when using Gricean Theory compared to Relevance Theory, and this trend was consistent even in the short prompts experiments. For *Metaphor*, models with 14B or more parameters tended to improve scores when using the proposed methods. In GPT-4o, GPT-4o-mini, and Qwen2.5-14B-Instruct, performance slightly improved with at least one of the proposed methods, even when using short prompts, though no performance improvement was observed with short prompts in phi-4. For *Deceits*, the effectiveness of the proposed methods varied depending on the model, and we observed no clear or consistent trend in relation to models' parameter size.

[3]For exact scores, see §I in the appendix.

4.2.3 Error Analysis

We conducted error analysis using the results from GPT-4o, categorized errors into the following five patterns, and counted the number of instances corresponding to each phenomenon (Figure 3). For ① and ③, actual examples of errors are presented in Table 1. Examples of the other error patterns are provided in Appendix Table 15.

① Cases where the proposed methods were clearly effective This category includes cases where `simple` and `cot` produced incorrect answers, but all other methods resulted in correct answers. Many problems classified under *Maxims* fell into this category. The first example in Table 1 corresponds to this pattern. We hypothesize that including pragmatic theories in the prompt made the model more sensitive to the distinction between *what is said* and *what is implied*.

② Cases where *Short Prompting* was insufficient This category includes cases where `grice` and `relevance` resulted in correct answers, but all other methods produced incorrect answers. A relatively large proportion of problems in this pattern fell under *Metaphor*. This suggests that understanding metaphors may not benefit from a superficial grasp of pragmatic theories alone, but a more detailed comprehension of these theories could enable their application to interpretation in such cases.

③ Cases where all methods failed In this category, problems classified under *Maxims* and *Irony* appeared in equal numbers. Since Figure 2 also indicates that these two phenomena were the most challenging for the models, it is natural that they appear prominently in this error pattern. Indeed, the second example in Table 1 seems difficult even for humans. Furthermore, determining that option 4 is the correct answer might require additional